

My project aims to analyze the [NREL Limited Access 2030 Moderate PV Supply Curve](#) dataset that is produced by NREL's reV (Renewable Energy Potential) Models. The Models produce a CSV file output that has a series of lat and long coordinates throughout the U.S. and analyzes every lat/long coordinates for it's solar site and development viability. The LCOE (Levelized Cost of Energy) LCOE describes the price per MWh a project(s) would need to sell electricity to cover all its costs over the project(s) lifetime. (The lower the cost the better). The goal is to identify patterns in solar development costs across California and evaluate which site characteristics most influence LCOE.

I will be focusing on supply curve metrics for Californias counties. I will be analyzing LCOE distribution, statistical profiling and performing Linear Regression and KNN ML modeling of LCOE outputs.

The project also integrates NoSQL storage via MongoDB and RapidMiner for visual machine learning workflows.